

AlgoProject — Capabilities, and Everything Already Built

What this is. A briefing for a reasoning agent that is helping design new robustness tests for this project but has only seen fragments of it. It answers three questions: **what this toolchain can actually do**, **what is already implemented** (so nothing is proposed twice), and **what is hard-blocked** (so nothing is proposed that cannot be built).

Written 2026-09-24 from the source tree, not from memory. Every empirical number carries the strategy, databank or feed it was measured on. In English on purpose — its reader is an agent, not the project's owner, whose own documents are in Spanish.

How to use it. Sections 1–3 are the machine and the pipeline. Section 4 is the inventory of what exists — skim the *question* column, not the file names. Section 5 is what is designed but unbuilt. **Sections 6 and 7 are the ones to read before proposing anything:** they are the constraints and the measured facts that kill most obvious ideas. Section 8 is an honest list of what nobody has covered.

1. The machine

1.1 Topology

StrategyQuant X (SQX) is a desktop application that generates and backtests strategies for MetaTrader 5. This project drives **three independent installs of it** on one 48-core machine:

install	role	port	who may use it
SQX (master)	the owner's GUI. Holds the real projects and the data store	—	the owner only
SQX_w1 (conductor)	headless worker for authoring, queries and short jobs	5060	agents, freely
SQX_w2 (custodian)	headless worker for the one long job at a time	5070	agents, freely, one job

The workers share the master's `user/data/History` by symlink. Python talks to a worker over an HTTP API (`sqcli` verbs) or reads its files directly with no SQX running.

1.2 What can be driven programmatically

□ Verified end to end:

- **Projects can be created, configured and run from code.** A project is cloned from a frozen donor, its tasks rewritten, its cross-checks enabled and configured, and the run launched.
- **Strategy files (`.sqx`) can be rewritten.** They are ZIPs; parameter values live in exactly one place, `strategy_Portfolio.xml` → `<variable><id>NAME</id>...<value>N</value>`, and the rules reference variables by name, so rewriting the value rewrites the rule.
- **Building blocks, random groups, strategy templates and build projects can be authored from a plain-English description** by agent skills, validated against the install's own vocabulary before import.
- **Databanks can be curated:** strategies removed by identity, so the next task in the chain sees only survivors.
- **Everything SQX produces can be read without SQX running** — see §1.4.

1.3 What cannot

- ⚠ **There is no import verb for price data.** `sqcli -data` only exports. Injecting a synthetic series (for a Masters-style permutation test) requires the master's GUI, which is the owner's.
- ⚠ **saveconfig output cannot be fed to loadconfig** — the verbs are asymmetric.
- ⚠ **A running instance must never have its `project.cfx` edited on disk:** SQX rewrites the file on save and exit and the change is lost silently.
- ⚠ **Every SQX sync deletes on-disk `.sqx` files not held in memory.** Anything that restarts SQX can destroy databanks.
- ⚠ **No external backtest engine exists.** `strategies/translate/` is empty. A strategy cannot be re-evaluated outside SQX, only its *trades* can be re-analysed.
- ⚠ **No `sqcli` verb changes a strategy's parameters.** The only way to score a chosen parameter combination is to write it into a `.sqx` and retest that file — which is exactly what the variant factory does.

1.4 What comes out, and what it costs to get

artefact	what it holds	cost
databank metrics export	41 aggregate metrics per strategy, with (IS) / (OOS) columns	seconds
trade export	every trade: entry/exit time and price, size, net P&L, MAE/MFE in account currency , sample tag, and exit reason (Close type)	~90 min for a few thousand strategies
dailyEquity.bin inside each retested .sqx	per-day P&L, per result leg and per cross-check market	□ 962 variants in 1.5 s, 5.9 MB of Parquet
SPP grid	~11–15k parameter permutations with full metrics per permutation	the SQX run
Walk-Forward Matrix	30 cells (steps × OOS share), per-step parameters and trades	the SQX run
Monte Carlo Retest	11 confidence levels per metric per task, plus per-simulation P&L vectors	the SQX run
M1 bar library	□ XAUUSD_DukasM1_Infinox = 7,949,285 bars, loaded in 0.4 s; every other timeframe is resampled from M1 and cached	trivial

Costs per asset (spread, commission, swap, slippage) are declared per symbol in versioned YAML and are **applied per task**, so a build and its out-of-sample retest carry different costs by design.

1.5 The variant factory — the capability worth understanding in detail

This is the piece that makes parameter-space work possible, and it is more capable than "generate variants" suggests.

1. **Reconnaissance.** An SPP grid (10–15k permutations) is read and reduced to a *design brief*: which parameters move the result, which are **provably inert** (the exact-duplicate test, not a variance threshold — see §7), each live parameter's marginal curve, its contiguous plateau and its centre.
2. **Design.** The brief becomes ~2–5k tuples in three strata: **neighbourhood** ($\pm k$ level steps around the plateau centre, complete), **coarse factorial** (a *complete* grid over fewer levels, saturated parameter by parameter in order of variance explained), and **coverage** (scrambled Sobol over every parameter, frozen ones included). Nothing here is a plain random draw.
3. **Controls.** Five rows exist to fail: the origin rebuilt through the factory, three known-result canaries taken from extreme SPP permutations, and one inert pair per frozen parameter. A broken chain returns a plausible middling number; extremes catch it.
4. **Fabrication.** Each tuple is written into a .sqx by textual substitution — byte-identical everywhere the tuple did not reach — with the identifier stamped **inside** the XML (SQX renames on collision) and the inherited <Fingerprint> removed (or the databank would deduplicate 5,000 variants into one).
5. **Verification.** The manifest reads the tuples back *off the files* and compares them against the plan. Tuple ≠ file is a silent failure in every other arrangement.
6. **Execution and harvest.** Loaded into the custodian, retested in three legs, then the metrics panel and the per-day equity curves are harvested.

There is a second, unrelated use of the same machinery: **period rescaling**, which fabricates siblings of one strategy with bar-unit parameters divided by a timeframe ratio, for the cross-timeframe test.

2. The data policy — and the one-way door

Every asset's history is cut into three segments, declared in versioned YAML:

segment	role
build	the only sample the generator ever sees
oos1	everything from the out-of-sample retest to the SPP: decay, Monte Carlo, MC Retest, cross-market, cross-timeframe, SPP
oos2	RESERVED for the walk-forward correlation and the walk-forward matrix — the last two tests

□ **oos2 is a one-way door and every look spends it.** Steps 7–16 read oos1 again and again, so by the time the final tests run it is exhausted; oos2 is the only untouched data left and it is fired once. Any proposed test must say which segment it reads, and a diagnostic that reads oos2 is almost always the wrong design.

A related rule: **the results of the last three tests are not looked at until all three have been run**, because seeing two of them contaminates the decision to run the third.

3. The pipeline — 21 steps

#	step	where
1	idea, in conversation	—
2	vocabulary: does the building block exist? if not, author and install it	agent skill
3	strategy template: fixed block + random hole, with a registry of what has been tried	agent skill
4	preflight: costs, windows and ranges — blocking	core/assets.py
5	create the custom project (cloned from a frozen donor)	sqx/projects/builder.py
6	configure the build	doctrine + YAML
7	out-of-sample retest in SQX	its own task, oos1 costs
8	IS/OOS screening in Python → apply verdict to the databank	gate/
9–10	cross-market retest in SQX, then its analysis	strategies/crossmarket/
10.5–12	rescaled siblings, cross-timeframe retest, then its analysis	strategies/crossTF/
13–14	Monte Carlo Retest in SQX (8 perturbation tasks), then its analysis	strategies/retest/
15–16	the two SPP runs, then the reconnaissance	strategies/sppUltra/
16.5	fabricate and retest the variant batch	sqx/variants/
17	walk-forward correlation	walkForwardCorrelation/
18	CSCV / PBO	walkForwardCorrelation/pbo.py
19	Walk Forward Matrix in SQX	strategies/walkForwardMatrix/
20	joint reading of 17, 18 and 19 — blind until all three exist	□ does not exist yet
21	exposure against buy-and-hold	strategies/exposure/

From 22 onwards the portfolio stage begins; it is barely started by design.

4. What is already implemented

Read the **question** column. The point of this section is that a proposal should either be absent from it, or be a sharper version of something in it with the difference stated.

4.1 Over a whole population (thousands of strategies at once)

module	the question it answers	how
gate/	of the thousands that were built, which few are worth spending machine time on?	a cascade of eight screens, each run over what the last left: presence (dropped by SQX = rejected), sanity , static metrics , degradation IS → OOS, shape , the monkey (§4.2), then two soft ones — family (Benjamini-Hochberg over the survivors) and redundancy (are these N strategies or one repeated N times, measured on equity curves). Joins the build and retest databanks on identity, emits a verdict SQX can apply
tasks/analysis/decay.py	how much of each strategy's in-sample edge survived, and is what is left bigger than its own error bar?	Sharpe either side, retention, Lo's standard error, years positive, share of profit from the best quarter → keep / doubt / discard
tasks/analysis/improvement.py	does filtering on an in-sample metric improve the out-of-sample outcome, and at what cost in survivors?	every metric swept at the 5/10/20/30/50 % cut from both ends, with a bootstrap interval on the difference; refuses to judge a filter leaving under 200 strategies
tasks/analysis/replication.py	does a conclusion drawn on one generation hold on another, independently generated one?	outcome gaps, thresholds carried across, rank stability
tasks/analysis/excess.py	over a family of tests, how many passed against how many chance alone would give?	excess, FDR, and whether the p-values are fine-grained enough to name anyone
tasks/analysis/correlations.py	which in-sample metric predicts which out-of-sample outcome?	Spearman-led, with the FDR correction across the family

4.2 One strategy, from its trade list

module	the question it answers	key design decisions
nulls/	how much of this result would a random trader with the same opportunity set have got?	A ladder of four rungs — randomise <i>when each trade is entered; when, and how long held; when, and the volatility sizing; free</i> (only trade count and cost survive). The gap between consecutive rungs is what that channel was worth. Costs are measured from the trades themselves; the fill convention is measured per strategy (□ open-to-open reconciles at 0.999985) and a p whose run did not reconcile is not read. A verifier proves null-against-null p-values are uniform before any real strategy is judged
strategies/monteCarlo/	given this trade history, what is the result's shape made of?	Five resampling models in two families — <code>iid_shuffle</code> , <code>block_shuffle</code> , <code>stationary</code> (reorder only) and <code>iid_bootstrap</code> , <code>block_bootstrap</code> (composition changes) — plus a stress family (missed entries, worse costs, fills degraded toward each trade's own MAE) and a regime family (volatility terciles by ATR or GARCH, calendar slices). Each model's table says what it preserves. Deliberately unseeded : stability across independent runs is measured instead
strategies/retest/	which single perturbed input breaks it?	Reads SQX's eight Monte Carlo Retest tasks: starting bar, spread, slippage, minimum distance, strategy parameters, exit parameters, OHLC, and all of them at once . Four questions in order: how much does it hurt (tail quantiles, conditional drawdown), how does it hurt (one regime or two), what breaks it (effect sizes in units of the control), do I believe it (multiplicity paid for)
strategies/crossmarket/	does the edge transfer to markets it never saw, or was it just being long?	Five placement nulls per market, each declaring what it randomises (block shift within regime and weekday-hour slot; segment permutation; resampled holds; fitted holds; regime strata). Re-derives the fill convention per market and refuses to interpret a market it could not reproduce . Carries a cost/execution stress with a breakeven cost multiple
strategies/crossTF/	does the edge survive being read on a slower clock?	Rescaled siblings run on the new timeframe and back on their own , the second being the control that separates "it died on H4" from "it died when its periods changed"
strategies/exposure/	what did that return cost in market time?	Occupancy, three buy-and-hold sizing conventions at matched risk, return per exposed hour, and how much of the market's move happened while holding
strategies/profitShape/	how few trades and how few periods does the result rest on; are the trades independent; did the mean change?	<code>S_top(1 %, 5 %)</code> , metrics with the best 5/10/20 trades removed, best-3-months and best-year shares, median vs mean · Wald–Wolfowitz runs, Ljung–Box on trades and on daily P&L, longest losing streak against 10,000 shuffles · OLS-CUSUM with both sides, rolling Sharpe with a non-normal band

module	the question it answers	key design decisions
strategies/entryQuality/	does the entry itself predict favourable movement, and what does arriving late cost?	MFE/MAE per horizon in ATR units off highs and lows, e(k) against a band of random entries matched on hour-of-day and long/short split , reported per direction; and the price given up by entering d bars (and d minutes) late, as a share of gross expectancy and as a multiple of modelled cost
nulls/filter.py	did a filter beat dropping the same share of trades at random?	Per-trade statistics, empirical p. Built, tested, and with no consumer yet — it waits for the ablation work in §5

4.3 One strategy, across a parameter space

module	the question it answers	key design decisions
strategies/sppUltra/	which parameters move the result, which are provably dead, and is this family worth 5,000 variants?	Eta-squared reported as a table over several metrics, never one column , plus the exact-duplicate test that decides freezing (§7). Contiguous plateaus and their centres. A noise check: the best point must clear what searching a pure-noise grid of the same effective size would produce
strategies/parameterCloud/	is the chosen point a lucky spike or a plateau; who moves the result; is there a surface at all; does its shape survive being cut into periods?	Rank, plateau fraction and shrunk expectation in a neighbourhood measured in level steps ; first-order and total Sobol indices computed on a quadratic surrogate; residual roughness <i>and</i> a model-free neighbour-disagreement roughness; gradient and Hessian at the origin; per-period profitable fraction, rank persistence, top-decile centroid and its drift; and an equal-weight blend of plateau members spread by farthest-point against the single point
walkForwardCorrelation/	does in-sample performance predict out-of-sample performance, and is the way <i>parameters are picked</i> prone to overfitting?	Rho with an interval that says indeciso rather than pretend; CSCV/PBO over C(12,6) = 924 partitions , run once per selection rule so the answer is "choosing by plateau centre instead of by maximum takes the PBO from 30 % to 4 %". Deflated Sharpe with an independent-trials count that refuses a cluster split leaving half the variants in one cluster
walkForwardMatrix/	does what optimises well predict what does well afterwards, or does re-optimising select for failure?	30 cells (steps × OOS share), rho per cell, optimum drift between steps, verdict predicts / blind / perverse

4.4 Shared statistical libraries

module	what it holds
core/surface/	dedupe : sentinel removal, distinct tuples, the <i>effective</i> n, and a bootstrap that resamples tuples rather than rows · shift : Hodges–Lehmann in units, Cliff's delta in rank, QQ curve, tail excess, dispersion ratio · plateau : plateau area, half-max area, the noise maximum, the deflated Sharpe, and the rank / plateau fraction / shrunk expectation of one chosen point
core/significance.py	Sharpe with its shape (skew, raw kurtosis), the variance factor both formulas share, the Probabilistic Sharpe Ratio, and the minimum track-record length
pipeline/	the chainer: mothers in, verdicts out days later; resumable, with a per-mother ledger and proof that deleting variants loses nothing
ledger/	the global search ledger : one appended line per search over a study's whole life, the funnel it produced, the map of which history has been spent and how often, the pooled N and sigma every deflated Sharpe should be eating, and the frozen thresholds. It also enforces the one-way door — a step that may not read a reserved segment is refused, and steps 17/18/19 are not served until all three have run
perf/	a cost catalogue: what every module costs in time, memory and disk, measured rather than estimated

5. Designed, specified, not yet built

Each of these is a self-contained written commission in `docs/encargos/`. They are **not** open questions — the design is settled; what stops them is named.

#	what it would add	what blocks it
8	Built on 2026-09-24 — see <code>ledger/</code> in §4.4. What remains of it: each module still reads its threshold from its own <code>config.yaml</code> , with <code>ledger/thresholds.yaml</code> as a register and a checker that refuses to let the two diverge	
9	A null population through the entire chain. 10,000 random-entry strategies pushed through all 21 steps, counting how many come out. "If 4 of 10,000 monkeys arrive and 5 of 10,000 of yours do, you know what that 5 is worth"	depends on 8. Must be done before the first real full pass, or the monkey count becomes a number compared against a known answer
10	Hansen's SPA and Romano–Wolf StepM over the surviving population: which strategies beat the benchmark once the whole search is discounted. Complementary to the CSCV, not a substitute — different matrix, different question	depends on 8
11	Edge per trade in cost units: gross expectancy per trade over mean spread, breakeven cost multiple, cost-to-edge ratio, net expectancy against a cost multiplier, broken down by entry hour	the percentage commission may be charged per leg or per trade — a factor of 2, unmeasured
12	Structural tests: rule ablation, signal inversion, and a seeded random-entry block inside SQX. The XML route is already investigated — ablation is deleting a <code><Block></code> from an <code><Item key="AND"></code> , and SQX declares <code>oppositeBlockKey</code> on custom blocks	needs strategy <i>logic</i> edited, not values; and D3 needs an integer hash inside a custom block, which the install's vocabulary may not provide
14	Conditional performance map: trades classified by volatility tercile, efficiency-ratio tercile, session and weekday	nothing technical — it is last on purpose, being the only test that <i>manufactures</i> hypotheses
15	A performance surface per market , and whether the good regions coincide across markets (Spearman and top-decile Jaccard)	costs unresolved for 16 assets, plus J × markets backtests of CPU
16	The trade replay simulator: re-execute each trade from a shifted entry along the M1 path, recomputing stops and sizing — the honest version of the delay test	the acceptance criterion needs a population with stops to validate against, and this one has none (§7)
17	Feed quality and bad-tick attribution: spikes, spike-and-revert, stale runs, gaps; then what share of profit sits within $\pm w$ bars of an anomaly	thresholds are the owner's to fix, in advance
6	Block taxonomy: labelling the 767 pooled building blocks by family	nobody has run it
13	Alpha/beta decomposition	parked by the owner , deliberately: the individual sequence closes first, and the factor that would matter most (returns of EAs already trading) does not exist yet

6. Hard constraints — read before proposing

Methodological, and not negotiable:

1. **Diagnosis, never selection.** Clouds, variants and clones are used to measure rank, plateau size, sensitivity, stability and transfer. Replacing the chosen strategy with the best-scoring clone produces a *more* overfit result with a smaller effective sample. Any parameter move is a human decision, logged, and revalidated on untouched data.
2. **No new filter comes out of a diagnostic** without being logged as a new trial and revalidated. With enough diagnostics, something always looks significant.
3. **Thresholds are frozen before the data is read.** A threshold set after seeing the numbers is not a threshold. Agents may read the threshold file, never edit it.
4. **oos2 is spent by looking at it (§2).**
5. **The owner's own SQX projects are his.** What a master project builds is a decision, never a bug to fix and never a finding to report.
6. **Costs are per task and declared per asset.** A test that quietly reuses one project's costs for another window is unattributable afterwards; every run happens in its own custom project.

Technical:

7. **The dependency set is small and pinned:** numpy, scipy, pandas, pyarrow, matplotlib, PyYAML, ruamel.yaml, arch, diptest, Flask, Markdown, and PySide6/FastAPI for the desktop app. **No scikit-learn, no statsmodels, no numba, no ruptures, no SALib.** Proposals that need one must say so and justify the install; anything reachable with scipy should use scipy.
8. **No import of synthetic price data (§1.3),** so bar-permutation tests in the Masters sense are not practicable today. The project's answer to that question is the monkey population (#9).
9. **One data vendor.** Dukascopy for everything but the Brent, so no cross-vendor data check is possible — which is exactly why #17 exists.
10. **One machine, and its limit is memory bandwidth, not cores.** □ Two AMD EPYC 7413 sockets: 96 logical threads but **48 physical cores**, and SMT *costs* throughput on this workload (211,419 paths/s on 48 physical against 203,617 on 96 logical). A compute-bound control that fits in L1 scales **46.1×** with 95 processes; the real out-of-cache kernel scales **18.3×** with the same 95 — the cores sit waiting for DRAM. Of 128 GB, the master JVM claims 71, the custodian is allowed 80, Python gets about 20 and the OS 10–12. **A proposal whose cost is "more simulations" is usually bounded by memory traffic**, and the fix that has worked here was restructuring the kernel to work in strips: 84× less memory and 3.6× faster at once. Several sessions share the two workers, so a long job is announced before it starts.

7. Measured facts that constrain any new design

These are all ☐ measured on this install, with the source named. Several of them have already killed an obvious-looking test.

About the parameter space

- **Two SPP runs cannot be paired.** The in-sample run and the out-of-sample run of one strategy, byte-identical settings, share **6 tuples out of ~11,600**. Nothing can compare two SPP windows directly — which is the entire reason the designed variant batch exists.
- **Eta-squared of a provably inert parameter is not zero — it is biased upward.**
`CBlock_SqzMmnInt21` gives 217 groups of tuples differing only in it and **all 217 produce an identical backtest**, yet it scores 0.0173, above any freezing threshold one would pick, because an SPP samples unbalanced. The converse also fails: a parameter scoring 0.0016 is demonstrably live. **Freezing is decided by the exact-duplicate test; variance explained only allocates levels.**
- **Rows are not observations.** Inert parameters duplicate points and the surface is smooth on top of that, with lag-1 autocorrelation up to **+0.89**; an interval over 12,000 correlated rows is falsely narrow by roughly a factor of three.
- **RExpectancy carries sentinels** — 99999.0 on 3 rows, -1.0 on 15 — which are 0.08 % of a grid **and win the argmax over 5,000 variants**.
- **The PBO of pure noise has a standard deviation of 0.21** over 252 partitions; individual draws ran 0.25 to 0.92 with a mean of 0.52. A grid at 0.45 and one at 0.55 are not distinguishable.
- **Regressing the chosen variant's out-of-sample Sharpe on its in-sample Sharpe measures a seesaw, not decay:** it reads -0.57 on pure noise and -0.99 on a panel with a genuine edge, i.e. *more* negative where the edge is real.
- **An unconstrained cluster count flatters the deflated Sharpe.** The silhouette prefers splitting 479 variants 478-against-1, which counts as two trials; refusing any split that leaves half the variants in one cluster gives 21 trials, and a deflated Sharpe of 0.66 instead of 0.91.

About this corpus of strategies

- The generated XAUUSD population carries **no stop, no target and no trailing**. ☐ Across 960,705 exported trades the exit reason takes exactly three values: `Exit After X Bars` (720,874), `Exit Signal` (187,853), `End Of Friday (Time)` (51,978).
- One studied strategy is **entirely long** — 1,119 of 1,119 trades.
- **Only 51.0 % of 757 strategies beat their entry-timing null on Sharpe, and 21.5 % on net profit.** The statistic moves the verdict more than the null does: real trades are **36 % less volatile** than random ones (551 \$ vs 720 \$ per trade, skew +0.54 vs -0.77, kurtosis 6.8 vs 27.1), and Sharpe rewards being calmer than chance while net profit does not.
- On that same corpus, **the e-ratio of a studied strategy sits inside the 5–95 % band of matched random entries at every horizon tested**, and below it at k=1. Its profit is 99 % from one year, and its best 5 % of trades carry 405 % of the total.
- **The CUSUM can call a series stable while its halves differ by a factor of sixteen** (+82.5 vs -5.1 per trade, supremum 1.01 against 1.36). At per-trade dispersion of hundreds of dollars the test has very little power.

- **Half a variant batch barely trades:** 1,002 of 2,000 rows under 30 trades, and filtering them out collapses one parameter to a single value — that parameter chooses between trading and not trading, not between good and bad.

About the data and the exports

- **1.2 % of XAUUSD entries do not land on a bar open** (far more on other projects): pending orders filled inside a bar, which is a price-conditional selection any bar-grid study must measure before trusting its comparison.
- **The fill convention is open-to-open**, measured per strategy at a correlation of 0.999985 against SQX's reported P&L; the runner-up convention reads 0.9629.
- **Costs are recoverable per trade** as `gross - reported P/L` — but $\triangle$ **the spread is inside the fill prices and does not appear in that residual**, so what is recovered is the commission (and swap overnight), not the spread.
- **A Monte Carlo Retest stores only a P&L vector per simulation** — 4 bytes per trade, no dates, no prices, no sizes, and **not the spread that was drawn**. Checked on 4,896 of 4,896 files.
- **A confidence level is an order statistic per metric, not a scenario.** The net profit and the drawdown on the "95 %" row come from different simulations; no row of that table is a coherent equity curve.
- **Names do not identify strategies:** two databanks of one project hold entirely different strategies under the same name, and 45 of 231 strategies had byte-identical trades under different hashes. Identity is the SHA-256 of the **normalised** inner XML — the raw hash matched 0 of 115 build-to-retest pairs, the normalised one matched 115 of 115.
- **The M1 library is clean where it is easy to check:** 0 OHLC inconsistencies in 7.9 M bars, but 40,097 flat bars (0.50 %) whose distribution nobody has looked at.

8. Where the holes are

An honest list, in case it suggests something.

1. **Step 20 does not exist.** There is no module that reads the walk-forward correlation, the CSCV and the walk-forward matrix *together* and issues one verdict — and by design none of the three may be looked at until all three have run.
2. **The global ledger now exists, and it is empty.** Only one study has been reconstructed into it (120 → 45 in eight screens) and, of its eight searches, **one** recorded the distribution of its candidates — a funnel stores counts, not scores. So the accumulated N is still the N of one gate run until the chain starts recording as it goes.
3. **The portfolio stage is barely started.** What is agreed: correlation between equity curves and not between metrics; drawdown computed on the aggregate, never summed; prop-firm rules are constraints, not after-the-fact filters. Everything else is open.
4. **Nothing measures redundancy against strategies already trading live**, because there are none yet. This is what parks the alpha/beta work.
5. **No test yet distinguishes "the entry is worthless" from "the exit is doing the work"** in a way that can be *acted* on — the e-ratio says the first, but the controlled comparison (the same exits with random entries, inside SQX) is commission #12 and unbuilt.
6. **Five analysis steps have no agent skill**, so the chain cannot run end to end unattended: cross-market (10), MC Retest (14), SPP reconnaissance (16), WFC (17) and CSCV (18).
7. **Costs are unresolved for 16 of 17 assets**, which blocks every multi-market design.
8. **Nothing checks the strategies against a market microstructure model** — no order-book assumptions, no partial fills, no queue position. Every fill in this project is an open price plus a modelled cost.

9. What a useful proposal looks like

Not a constraint on creativity — a way of making an idea checkable against everything above. A proposal that answers these seven lines can be costed and built; one that does not usually turns out to be item 4.2 in a new hat, or to need data that cannot be imported.

the question	in one sentence, and it must not already be in §4 — or, if it sharpens one, say what the difference buys
unit of observation	a trade, a bar, a strategy, a parameter tuple, a market, a period. This decides everything else
input	which artefact of §1.4, and whether it already exists or needs an SQX run
segment	<code>build</code> , <code>oos1</code> , or — with an explicit justification — <code>oos2</code>
cost	Python seconds, or SQX hours; if the latter, why it is worth a custodian slot
diagnostic or gate	and if a gate, what the threshold is and why it can be frozen before looking
failure mode	what result would make the test <i>wrong</i> rather than merely negative. A test nothing can falsify measures nothing

Two habits of this project worth borrowing in any proposal:

Reconcile before interpreting. Every module that prices something re-derives what SQX already computed and refuses to speak if the two disagree — the fill convention at 0.999985, the canaries in a variant batch, the manifest read back off the files. Every one of those checks has caught a real failure that produced plausible numbers.

Say what the null holds fixed. Every randomisation in this codebase carries a table of what it preserves and what it hands to chance, printed next to the numbers it produced. A null that changes two things at once cannot attribute a result to either.